

Reproducing the Baseline Results for the POLY-SIM Grand Challenge

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Abstract

This report reproduces the baseline results for the POLY-SIM Grand Challenge, a closed-set speaker classification task that uses both face and voice information. The experiments evaluate multimodal recognition under same-language, cross-lingual, and missing-modality conditions. The provided FOP baseline was trained on the v3 split with German as the seen language and English as the unseen language. The reproduction confirms that multimodal evaluation gives the strongest performance, while removing the face modality leads to a clear drop in accuracy. A comparison between two loss settings also showed that the plain cross-entropy configuration outperformed the setting with orthogonal projection loss in this setup.

1. Introduction

The POLY-SIM challenge studies speaker identification with two modalities: voice and face. Unlike a standard multimodal classification task, the benchmark also tests whether the model remains effective when one modality is missing at test time and when the test language differs from the training language. This makes the task more realistic and more difficult than standard in-language multimodal recognition. The objective of this assignment was to reproduce the provided baseline and report the obtained results. The baseline code and embeddings were supplied in the repository, so the main focus was understanding the experimental setup, running the training pipeline, verifying the evaluation protocol, and comparing the main task settings required by the challenge.

2. Task Definition

The challenge contains four important settings. P3 is the in-language multimodal case, where both voice and face are available during testing and the test language matches the training language. P4 is the missing-modality case, where the model is still trained with both modalities but is tested with audio only. P5 is the cross-lingual multimodal case, where both modalities are available but the test language is different. P6 is the hardest setting, combining cross-lingual testing with missing face information at inference time. All experiments use closed-set speaker classification. Therefore, each test sample must be assigned to one of the known speaker identities present in the benchmark split. The primary evaluation metric is **classification accuracy**.

3. Baseline Method

The provided baseline is the FOP model. It contains one branch for face embeddings and one branch for voice embeddings. Instead of learning directly from raw images and raw waveforms, the implementation uses pre-extracted features stored in NumPy files. This simplifies the pipeline and allows the experiments to focus on multimodal fusion and classification. Each branch projects its input embedding into a learned representation,

and the two branches are then combined by a linear fusion module. The fused representation is passed to a classifier that predicts the speaker identity. In the used split, the model predicts one of 36 classes.

Two loss settings were tested. The first setting used only **cross-entropy loss**, corresponding to $\alpha = 0.0$. The second setting **combined cross-entropy** with orthogonal projection loss, using $\alpha = 1.0$. The second option was included because the original baseline paper emphasizes the role of orthogonal constraints in improving representation quality.

4. Implementation Details

The project was executed in Visual Studio Code on a local Windows machine. The original repository recommends Python 3.6.5 and PyTorch 1.8.0, but the code ran successfully on a newer local environment after making a few compatibility corrections. First, the device setting was changed from CUDA to CPU because the installed PyTorch build did not include CUDA support on the local machine. Second, the path in the configuration file was updated to match the local project directory. A small correction was also required in the test script. The original sweep function created masked face and audio tensors, but the evaluation call did not actually use the masked face tensor. This was corrected so that the missing-modality evaluation truly reflected the intended audio-only protocol when face inputs were removed at test time.

The baseline training was run on version v3 with German as the seen language and English as the unseen language. The code was executed twice for the loss comparison: once with $\alpha = 0.0$ and once with $\alpha = 1.0$. After training, the saved best checkpoint from the stronger run was used for missing-face evaluation. Code accompanying this report is provided separately as a repository, as required by the assignment.

5. Experimental Setup

The training split was loaded from the provided CSV file for German. The seen test set used the German test split, while the unseen test set used the English split. The main training script reported accuracy after each epoch for both seen and unseen evaluation. Early stopping was triggered automatically based on validation behavior.

To simulate missing modality, the test script was used to progressively set a selected percentage of face embeddings to zero. The final challenge settings P4 and P6 correspond to the extreme case of 100% missing face, because the test input then contains audio only. This matches the task definition provided in the challenge description.

6. Results

Setting	Description	Accuracy
P3	In-language multimodal (German, audio + face)	93.86%
P4	Missing-modality / audio only (German)	76.90%
P5	Cross-lingual multimodal (English test, audio + face)	91.38%
P6	Cross-lingual missing-modality / audio only	74.57%

Table 1. Main reproduced results for the four challenge settings.

The reproduced results show a very clear trend. When both modalities are available, the model performs strongly in both same-language and cross-lingual evaluation. P3 reached 93.86%, while P5 reached 91.38%.

This indicates that the baseline generalizes reasonably well across languages when it can still rely on both face and voice information.

When the face modality is completely removed, performance drops noticeably. In the same-language audio-only condition, accuracy falls to 76.90%. In the cross-lingual audio-only condition, the score decreases further to 74.57%. This confirms that the visual stream is useful for identity discrimination and that cross-lingual robustness remains more challenging when only speech is available.

7. Loss Comparison

The comparison between $\alpha = 0.0$ and $\alpha = 1.0$ also produced an important observation. The run with $\alpha = 0.0$ achieved the best seen accuracy of 93.86% and the best unseen accuracy of 91.38%. In contrast, the run with $\alpha = 1.0$ achieved 89.77% on the seen split and 90.09% on the unseen split. Therefore, in this reproduction, adding the orthogonal projection loss did not improve the final results.

One possible interpretation is that the provided embeddings already contain strong identity information, so the additional orthogonality constraint may not have been beneficial in this exact training setup. Another possibility is that the best value of α is task-dependent and may require more tuning than was performed in this assignment.

8. Discussion

The report reproduces the core behavior expected from the POLY-SIM challenge. Multimodal testing leads to the highest accuracy, while removing one modality makes the task significantly harder. Moreover, the effect of language mismatch is relatively small in the multimodal case but becomes more visible in the audio-only setting. This suggests that the face branch provides stable complementary information that helps compensate for language differences.

From an implementation perspective, the provided codebase was simple enough to run after minor adjustments. The most important step was verifying that the evaluation script really applied the intended missing-modality masking. Without this correction, the reported missing-modality scores would not have matched the challenge definition accurately.

9. Conclusion

The baseline for the POLY-SIM Grand Challenge was successfully reproduced. The strongest results were obtained with $\alpha = 0.0$, achieving 93.86% for P3 and 91.38% for P5. When the face modality was completely removed, the model achieved 76.90% for P4 and 74.57% for P6. Overall, the experiments confirm that multimodal fusion is highly effective, but missing-modality and cross-lingual conditions still reduce performance. Future improvements could focus on stronger modality-agnostic fusion and more robust speech representations for cross-lingual audio-only classification.

10. References & repository

- [1] M. S. Saeed, M. H. Khan, S. Nawaz, M. H. Yousaf, and A. Del Bue, "Fusion and orthogonal projection for improved face-voice association," in ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 7057-7061, 2022.
- [2] POLY-SIM Grand Challenge assignment materials and provided baseline repository.

The obtained results are internally consistent across all four evaluation conditions. In particular, the ranking $P3 > P5 > P4 > P6$ is meaningful because it reflects the increasing difficulty of the challenge settings.

Although the absolute scores are strong, the large gap between multimodal and audio-only conditions indicates that the model still relies substantially on the face representation for correct speaker identification.

This behavior is expected in speaker identification with paired face-voice embeddings, because the two modalities carry complementary identity cues. When one cue is removed, the classifier must make a decision from a weaker signal.

For future work, a more advanced fusion strategy, stronger regularization, or more balanced multilingual training data may further improve the most challenging setting.